Zuu Crew
Machine Learning
Acadamy

FINAL ASSESSMENT

PulseTrack

Data Warehousing Capstone Project

Final Assessment - Lead Data Engineer / Data Architect Level

Objective: Design and implement one complete analytical data platform from the supplied PulseTrack source data through an Enterprise Data Warehouse, Data Mart and OLAP/Semantic layer.

Assessment	Data Warehousing Capstone
Level	Lead Data Engineer / Data Architect
Source Data	CSV, JSON and PostgreSQL
Technology	Student-selected and justified
Core Requirement	One integrated end-to-end solution

Important: This is one continuous capstone solution. Each question produces an output that must be used by the following questions. Students must not create unrelated solutions for individual questions.

Project Overview

PulseTrack is a performance-marketing platform connecting advertisers, campaigns, offers, creatives, partners and customer conversion activity.

Students are appointed as Lead Data Engineers responsible for designing and implementing an analytical platform from the supplied source data. The source environment contains CSV, JSON and PostgreSQL data and includes data-quality, relationship, historical and processing challenges.

No specific ETL, ELT or orchestration technology is prescribed. Students must select appropriate technologies and defend their decisions.

The final solution must evolve as one integrated platform:

Source -> Data Platform -> Data Warehouse -> Data Mart -> OLAP/Semantic Layer -> Business Analytics

Assessment Principles

- Business requirements must drive the warehouse design.
- Source analysis must influence architectural decisions.
- The dimensional model created in Question 2 is the target model used throughout implementation.
- Historical business context must be preserved.
- ETL/ELT and orchestration technologies must be selected and justified rather than prescribed.
- The Data Mart must be built from the student's Enterprise Data Warehouse.
- The OLAP/Semantic layer must be built from the student's Data Mart.
- Final analytical results must be traceable back to source data.
- Changes to earlier design decisions must be documented and justified.

Question 1 - Business KPIs, Analytical Requirements & Grain (15 Marks)

PulseTrack management wants to measure the performance of its performance-marketing platform.

Required analytical areas

- Advertiser Performance: impressions, clicks, CTR, conversions, conversion rate, revenue, revenue per conversion and revenue per click.
- Campaign Performance: impressions, clicks, conversions, conversion rate, revenue, adjusted revenue and campaign ranking.
- Creative Performance: impressions, clicks, CTR, conversions and conversion rate.
- Partner Performance: clicks, conversions, attributed conversions, attributed revenue and attribution percentage.
- Account Manager Performance: conversions, revenue, advertiser performance and campaign performance.
- Time Analysis: year, quarter, month and day.

Management also requires month-over-month, quarter-over-quarter, year-over-year, Top-N, percentage-contribution and trend analysis.

Historical requirement: all historical reports must represent the business context that was valid when the event occurred. For example, if an advertiser changes account manager, historical conversions must remain associated with the account manager responsible at the time of the conversion.

Student Task

1. Define the business processes represented by the requirements.
2. Define the KPIs and measures.
3. Identify the required dimensions.
4. Determine the required grain for each analytical process.
5. Identify historical requirements.
6. Define the analytical questions that the final platform must answer.
7. Define reasonable data-freshness and retention requirements.
8. Map each KPI to its required measures, dimensions, grain and source data.

Deliverable: PulseTrack Analytical Requirements & KPI Specification. This becomes the foundation for all subsequent questions.

Question 2 - Source Forensics & Enterprise Data Warehouse Design (20 Marks)

Using the supplied PulseTrack EER and source datasets, perform a source-system forensic analysis and design the Enterprise Data Warehouse required by Question 1.

Source analysis

- Analyse source formats, table grain, business keys, candidate keys and relationships.
- Identify duplicate records, orphan records, denormalised information, M:N relationships, late-arriving data and historical information.
- Investigate the supplied source issues, including duplicate conversion identifiers, nullable/orphan click relationships, embedded advertiser/campaign/partner names, partner identifier relationships, conversion touchpoints, revenue adjustments and account-manager-at-event history.
- Explain why directly joining the RAW source tables into a reporting dataset can produce incorrect results, including fanout and grain-mismatch problems.

Warehouse design

- Determine the business processes represented by the KPI requirements.
- Design the required Fact tables, Dimension tables and Bridge tables where justified.
- State the grain of every Fact table explicitly.
- Define measures and classify their additive behaviour.
- Define surrogate keys and natural/business keys.
- Identify conformed dimensions.
- Use degenerate dimensions where appropriate.
- Explain how the model supports every KPI defined in Question 1.

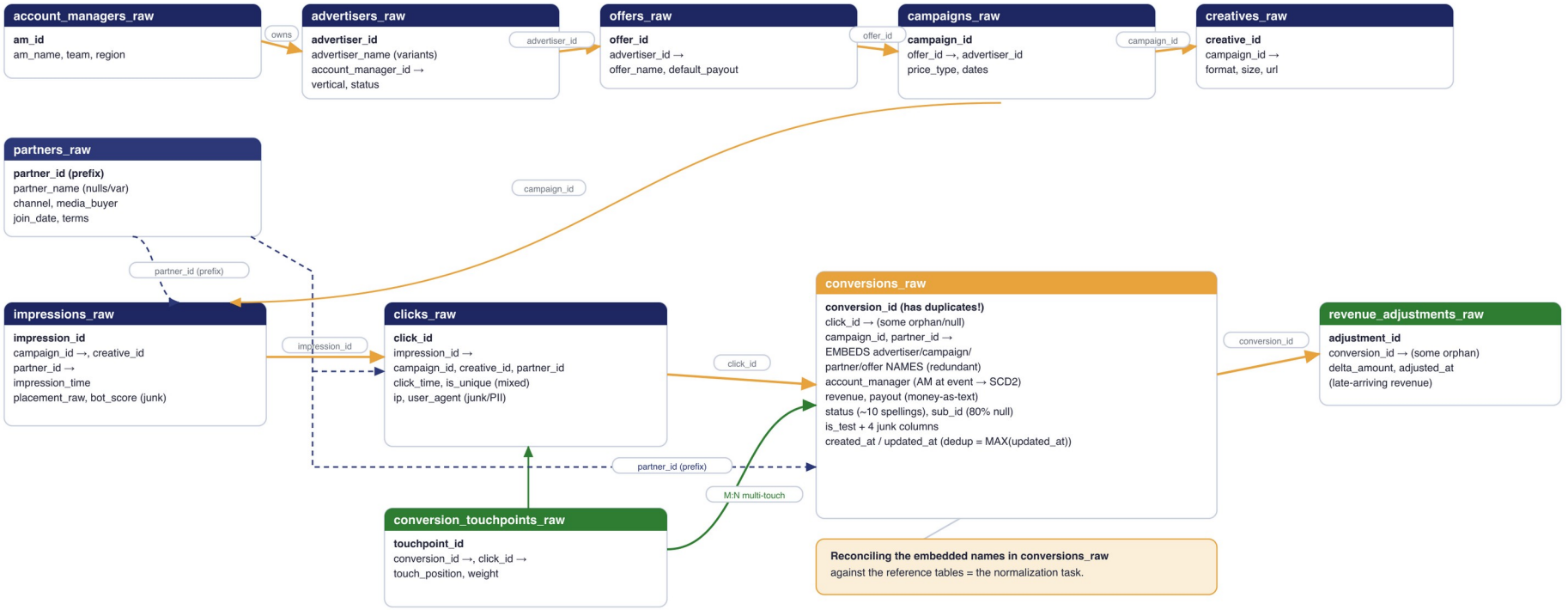
History preservation

The warehouse must preserve historical business context. Design and implement the appropriate historical strategy for dimensions requiring history, including the account-manager-at-event requirement. Demonstrate how historical events reference the correct dimension version.

> **Deliverable: Final PulseTrack Enterprise Data Warehouse dimensional model. This becomes the single target model used by every subsequent question.**

PulseTrack — RAW sources (Bronze) · 11 relational tables + MongoDB + event stream

one platform, three sources, one ground truth (they reconcile). Every table carries created_at / updated_at audit columns. Header fills: gold = centrepiece/keys, green = companion & late-arriving, navy = platform. Gold = key link · dashed = prefix (soft) link · green = bridge/M:N edges (arrows point to the joined tabl



MongoDB — PartnerHub / CRM (mongo/*.jsonl)

partner_profiles (600 docs)
nested contact{} / channels[] / payment{} / tags[]
overlaps partners_raw on the id prefix (~12% name conflicts)
40 portal-only docs RESOLVE the orphan prefixes
advertiser_accounts (60 docs)
account_manager_history[] = SCD2 source of truth

Event stream — Kafka-style (stream/*.log)

topic pulsetrack.conversions / 4 partitions
conversion.created / status_changed / adjustment.applied
at-least-once: ~2.5% duplicate deliveries (dedup by event_id)
~0.1% malformed lines / schema v2 cutover at ~70%
replaying the stream == batch final state
(conversions MAX(updated_at) + adjustments)

Cross-source keys (they reconcile)

partner prefix <-> partner_profiles.partner_id
advertiser_id <-> advertiser_accounts.advertiser_id
conversion_id <-> stream key / event payloads
account_manager_history[] validates embedded AM (SCD2)
stream replay validates dedup + adjustments
created_at / updated_at -> incremental watermarks

Question 3 - Data Platform Architecture & ETL/ELT Tool Selection (15 Marks)

Design the target data platform and select the technologies required to implement it. No ETL, ELT or orchestration technology is prescribed.

Architecture

Your architecture must show the complete lifecycle from sources through ingestion, data processing, the Enterprise Data Warehouse, Data Mart and OLAP/Semantic layer. It must also address orchestration, metadata, audit, monitoring, error handling, retry, recovery, lineage and security.

Technology selection

Evaluate at least three possible approaches. Consider, where relevant, tools such as Apache NiFi, Apache Airflow, dbt, Apache Spark, Python, SQL, cloud-native services or other appropriate technologies.

- CSV ingestion
- JSON ingestion
- PostgreSQL extraction
- Transformation capability
- Large-volume processing
- Scheduling
- Dependency management
- Incremental processing
- Retry and recovery
- Backfill
- Data quality
- Maintainability
- Scalability
- Monitoring

Use a consistent scoring and weighting approach. Do not justify a technology simply because it is popular or familiar.

Final decision: select and justify the ingestion, transformation, orchestration, warehouse and analytical-serving technologies. Clearly define the responsibility of each selected technology.

Deliverable: Technology Decision Record + Final Data Platform Architecture.

Question 4 - Implement the ETL/ELT Pipeline (20 Marks)

Using the architecture and technology decisions from Question 3, implement the warehouse designed in Question 2.

The implementation must demonstrate

- CSV ingestion
- JSON ingestion
- PostgreSQL extraction
- Bronze/raw preservation with ingestion metadata
- Silver cleansing and standardisation
- Duplicate handling
- Key reconciliation
- Relationship validation
- Business-rule validation
- Historical/SCD processing
- Dimension loading
- Fact loading
- Bridge loading where required
- Referential integrity
- Correct Fact grain

Important: The implementation must use the warehouse design from Question 2. If a design problem is discovered, document the change, justify it and explain its downstream impact.

Question 5 - Incremental Processing, Idempotency, Late-Arriving Data & Recovery (10 Marks)

Design and implement the processing strategy for repeated, changed and late-arriving source data.

The solution must handle

- new records
- changed records
- duplicate files
- duplicate events
- late-arriving events
- late revenue adjustments
- historical corrections
- backfills

Required scenarios

- Duplicate delivery: the same 50,000 records are delivered twice. The expected analytical result is 50,000 valid records, not 100,000.
- Late revenue: a conversion occurs on 1 August and its revenue adjustment arrives on 5 August. Update the analytical result without creating another conversion.
- Historical change: an advertiser's account manager changes. Historical conversions continue to reference the correct historical dimension version.
- Pipeline restart: the pipeline fails after loading dimensions but before completing fact loading. Demonstrate safe recovery.

Question 6 - Data Quality, Reconciliation & Audit (10 Marks)

Make the implemented warehouse a trusted analytical system.

Quality checks

- schema, null, data-type and malformed-record checks
- duplicate detection
- referential integrity across advertiser, offer, campaign, creative, impression, click, conversion, partner, touchpoint and revenue-adjustment relationships
- valid dates, monetary values and statuses
- business-rule and event-sequence validation
- attribution validation

Reconciliation

Reconcile Source -> Bronze -> Silver -> Gold, including source count, valid count, rejected count, duplicate count, inserted count and updated count.

Audit

Implement an audit mechanism containing appropriate information such as batch ID, pipeline ID, run ID, start time, end time, status, source record count, processed count, inserted count, updated count, rejected count and error count.

Question 7 - Data Mart Design & Implementation (10 Marks)

The Enterprise Data Warehouse is now complete. Build a business-oriented Data Mart from your own warehouse that supports all KPIs defined in Question 1.

- Determine the Data Mart grain.
- Select the required facts and dimensions.
- Define measures and calculated metrics.
- Define required aggregations.
- Explain why the Data Mart is required rather than allowing every BI query to directly access the detailed warehouse.
- Implement and populate the Data Mart from the final Enterprise Data Warehouse.

Deliverable: Working PulseTrack Data Mart.

Question 8 - Advanced OLAP Cube / Semantic Analytical Model (15 Marks)

Build an advanced OLAP Cube or Semantic Analytical Model using the Data Mart from Question 7.

Dimensions and hierarchies

- Date: Year -> Quarter -> Month -> Day
- Advertiser: Advertiser -> Campaign -> Offer
- Campaign: Campaign -> Creative
- Partner: appropriate business hierarchy based on the source model and justified assumptions

Measures

- impressions
- clicks
- conversions
- revenue
- adjusted revenue
- attributed revenue
- CTR
- conversion rate
- revenue per click
- revenue per conversion
- attribution percentage
- contribution percentage

Question 8 - Advanced OLAP Cube / Semantic Analytical Model (continued) (15 Marks)

OLAP operations

- roll-up
- drill-down
- slice
- dice
- pivot
- Top-N
- ranking
- percentage contribution
- MoM
- QoQ
- YoY

Advanced requirements

- Classify measures as additive, semi-additive or non-additive.
- Define calculated measures and KPIs.
- Design meaningful hierarchies and levels.
- Explain aggregation strategy.
- Address high-cardinality dimensions.
- Address cube/semantic-model performance.
- Address incremental processing and pre-aggregation/caching where appropriate.

Question 9 - Business Analytics Using the Final Platform (10 Marks)

Using only the final analytical platform, answer complex business questions. Do not bypass the Data Warehouse/Data Mart/OLAP architecture by querying RAW data directly.

- Which advertisers generated the highest revenue each quarter?
- Which campaigns have the highest CTR but the lowest conversion rate?
- Which partners generated the highest attributed revenue?
- How did revenue change compared with the previous month?
- Which campaigns contributed more than 10% of total revenue?
- Which creatives underperformed relative to their campaign?
- How did an advertiser's performance change before and after an account-manager change?
- What percentage of total revenue is attributable to each partner?
- Which campaigns show increasing impressions but declining conversions?
- What is the revenue impact of late adjustments?

For each result, demonstrate traceability from Business KPI -> OLAP/Semantic Measure -> Data Mart -> Gold Fact/Dimension -> Silver -> Bronze -> Source.

Question 10 - Production Architecture & Technical Defence (5 Marks)

Defend the complete solution as a Lead Data Engineer. Explain how the architecture behaves under the following scenarios:

- Source volume increases by 10x.
- A source JSON schema changes.
- The same file is delivered three times.
- Revenue adjustments arrive seven days late.
- A historical account-manager change is received late.
- The ETL process fails halfway through.
- A BI user requests five years of campaign-level analysis.
- A business user claims warehouse revenue differs from the source system.

For each scenario, explain how the existing architecture handles the situation, what component is responsible, and what operational evidence would be used to investigate or resolve the issue.

Final Capstone Solution

The questions are intentionally dependent. The final submission must represent one coherent solution:

Business KPIs & Requirements -> Source Forensics -> Enterprise Data Warehouse -> Technology & Architecture -> ETL/ELT Implementation -> Incremental/Idempotent Processing -> Data Quality & Audit -> Data Mart -> OLAP/Semantic Layer -> Business Analytics -> Production Defence

Required Final Deliverables

- Analytical Requirements & KPI Specification
- Source/EER Forensic Analysis
- Technology Decision Record
- End-to-End Data Platform Architecture
- Enterprise Dimensional Model / Star Schemas
- Historical/SCD design and implementation evidence
- Working ETL/ELT pipeline
- Working Enterprise Data Warehouse
- Incremental and idempotent processing evidence
- Data-quality and reconciliation evidence
- Audit mechanism and operational evidence
- Working Data Mart
- OLAP Cube / Semantic Analytical Model
- Business analytical results
- Final architecture and technical defence

Assessment Rule

The final solution must be internally consistent. Decisions made in earlier questions must flow into later implementation and analytical layers. If a student changes a design decision, the change and its impact must be documented and justified.

The objective is not to demonstrate knowledge of individual tools in isolation. The objective is to demonstrate the ability to design, implement, operate and defend a reliable enterprise data warehouse as a Lead Data Engineer.

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